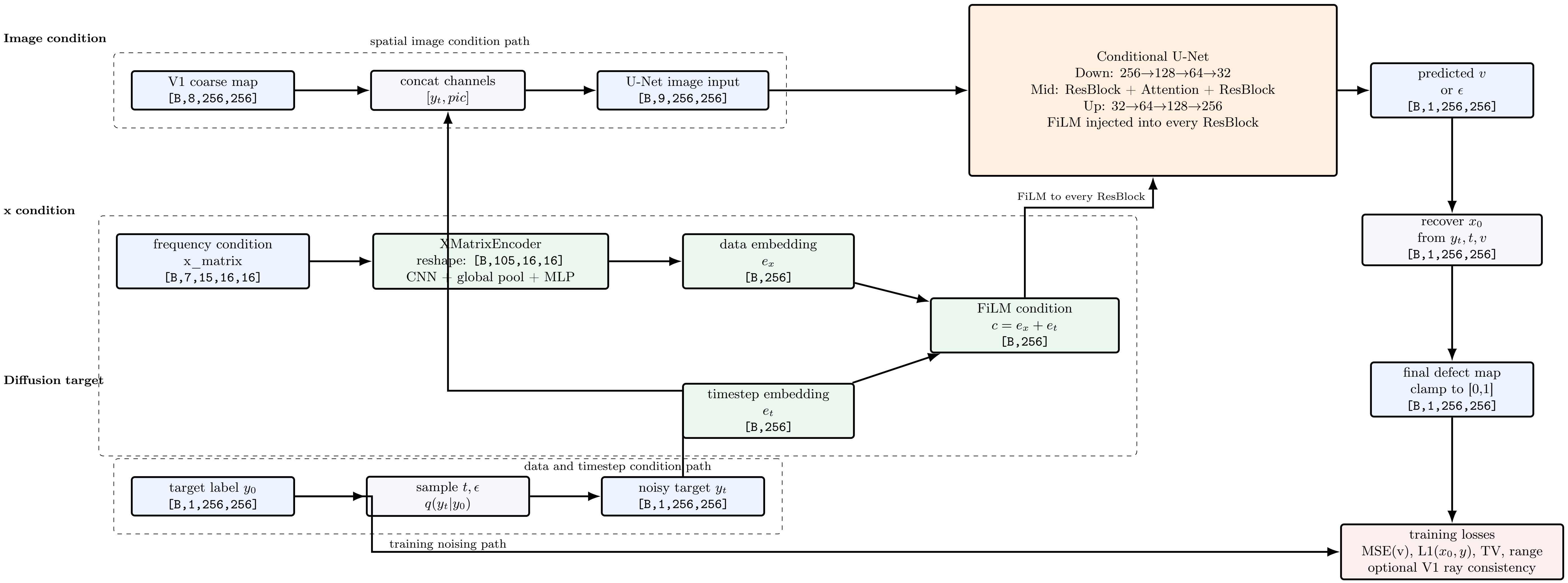
Conditional Diffusion Network for Ultrasonic Defect Map Inversion

Main drawing follows `dataset_a_256_base48.yaml`: image size 256, pic channels 8, x channels 7, frequencies 15, base channels 48.
Tensor order: image tensors use `[B,C,H,W]`; x-matrix uses `[B,C,F,TX,RX]`.

Training view. Diffusion variable is the continuous defect-depth label y, not the coarse map.
The coarse map gives spatial prior; x_matrix preserves tx-rx-frequency information.

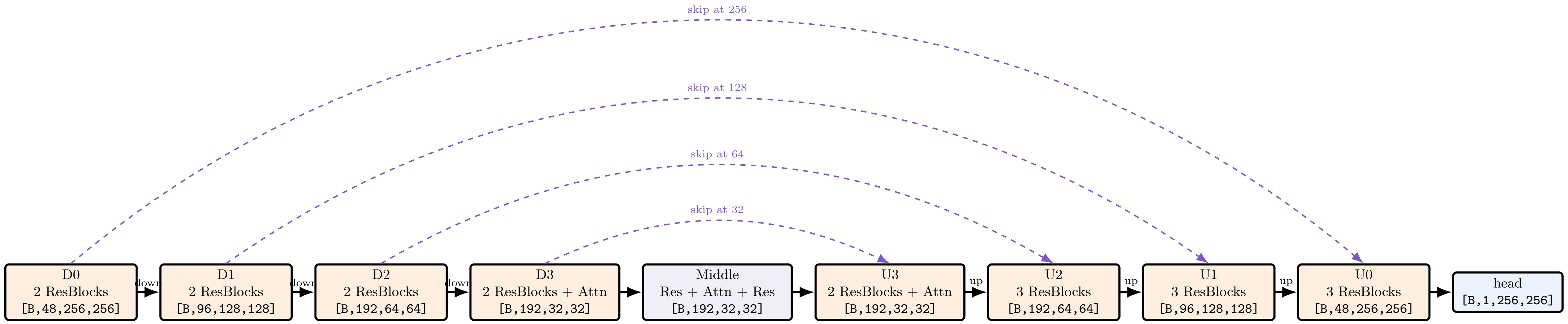
Sampling view. Remove y_0 and loss nodes. Start from Gaussian noise `[B,1,256,256]`; run DDIM/DDPM denoising with the same pic and x_matrix conditions.



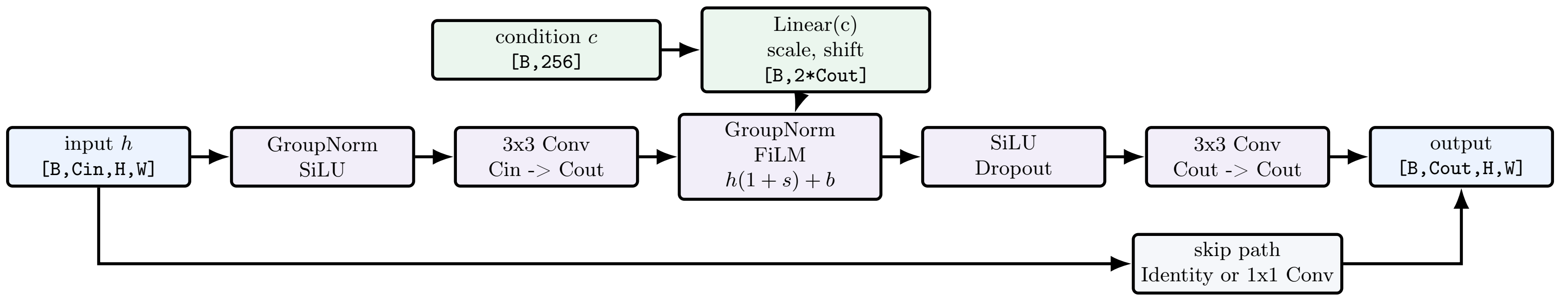
U-Net Tensor Shapes and Block Counts

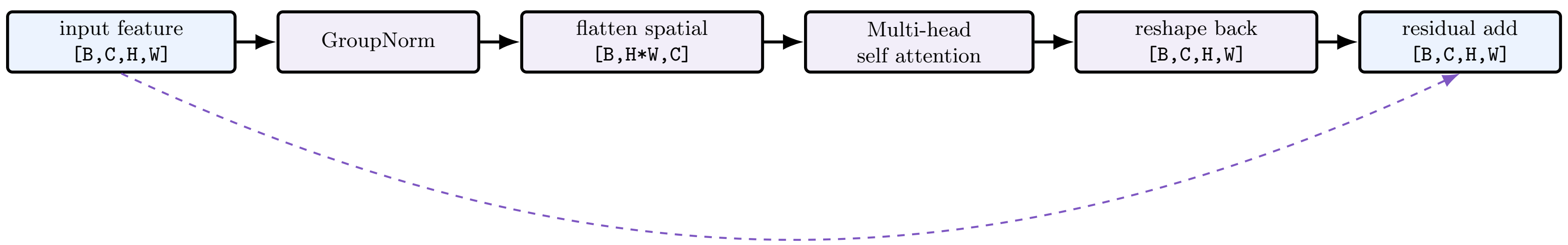
Configuration: base=48, channel_mult=[1,2,4,4], num_res_blocks=2, attention_resolutions=[32].

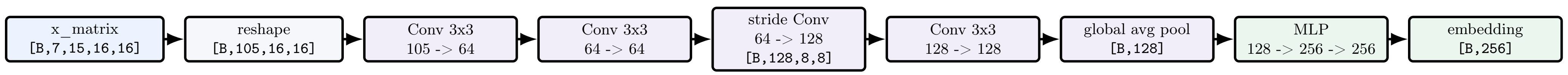
Stage	Operation	Input shape	Output shape	Notes
Input concat	concat y_t and pic	`[B,1,256,256]` + `[B,8,256,256]`	`[B,9,256,256]`	image condition is channel-concatenated.
Input conv	3x3 conv	`[B,9,256,256]`	`[B,48,256,256]`	first feature map.
Down level 0	2 x ResBlock(48)	`[B,48,256,256]`	`[B,48,256,256]`	stores 2 skip tensors at 256.
Downsample 0	stride-2 conv	`[B,48,256,256]`	`[B,48,128,128]`	stores pre-downsample skip.
Down level 1	ResBlock 48->96, ResBlock 96->96	`[B,48,128,128]`	`[B,96,128,128]`	stores 2 skip tensors at 128.
Downsample 1	stride-2 conv	`[B,96,128,128]`	`[B,96,64,64]`	stores pre-downsample skip.
Down level 2	ResBlock 96->192, ResBlock 192->192	`[B,96,64,64]`	`[B,192,64,64]`	stores 2 skip tensors at 64.
Downsample 2	stride-2 conv	`[B,192,64,64]`	`[B,192,32,32]`	stores pre-downsample skip.
Down level 3	2 x ResBlock(192) + Attention after each	`[B,192,32,32]`	`[B,192,32,32]`	stores 2 skip tensors at 32.
Middle	ResBlock + Attention + ResBlock	`[B,192,32,32]`	`[B,192,32,32]`	bottleneck.
Up level 3	2 x concat skip + ResBlock(384->192) + Attention	`[B,192,32,32]`	`[B,192,32,32]`	consumes two 32 skips.
Upsample 3	nearest upsample + 3x3 conv	`[B,192,32,32]`	`[B,192,64,64]`	moves to 64.
Up level 2	3 x concat skip + ResBlock	`[B,192,64,64]`	`[B,192,64,64]`	consumes downsample skip + two 64 skips.
Upsample 2	nearest upsample + 3x3 conv	`[B,192,64,64]`	`[B,192,128,128]`	moves to 128.
Up level 1	3 x concat skip + ResBlock	`[B,192,128,128]`	`[B,96,128,128]`	final channel count at this level is 96.
Upsample 1	nearest upsample + 3x3 conv	`[B,96,128,128]`	`[B,96,256,256]`	moves to 256.
Up level 0	3 x concat skip + ResBlock	`[B,96,256,256]`	`[B,48,256,256]`	final high-res feature map.
Output head	GroupNorm + SiLU + 3x3 conv	`[B,48,256,256]`	`[B,1,256,256]`	predicts v by default.



Block Internal Structure







Block	Stack rule in code
ResBlock	GroupNorm + SiLU + Conv, then FiLM-conditioned GroupNorm, SiLU + Dropout + Conv, plus identity or 1x1 skip. Every ResBlock receives the same condition vector $c = e_x + e_t$.
AttentionBlock	Used only where spatial resolution equals 32 for base48 config. It is self-attention over the flattened $H \times W$ locations.
Downsample	3x3 stride-2 convolution. Channels stay the same; resolution halves.
Upsample	nearest-neighbor upsample by 2, then 3x3 convolution. Channels stay the same; resolution doubles.
Skip concatenation	Decoder ResBlocks concatenate the current feature with the latest saved encoder skip tensor along the channel dimension before the ResBlock.

Step	Formula and tensor shape
Forward noising	$y_t = \sqrt{\bar{\alpha}_t} y_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$, all image tensors are `[B,1,256,256]`.
Condition image	input = concat(y_t, pic) gives `[B,9,256,256]`.
Condition vector	e_x = XMatrixEncoder(x), e_t = TimeEmbedding(t), $c = e_x + e_t$, all `[B,256]`.
Network prediction	$\hat{v}$ = ConditionalUNet$([y_t, pic], t, e_x)$ with shape `[B,1,256,256]`.
Training target	With `prediction_type=v_prediction`: $v = \sqrt{\bar{\alpha}_t}\epsilon - \sqrt{1-\bar{\alpha}_t} y_0$.
Main loss	L_{diff} = MSE$(\hat{v}, v)$.
Auxiliary clean prediction	$\hat{x}_0 = \sqrt{\bar{\alpha}_t} y_t - \sqrt{1-\bar{\alpha}_t}\hat{v}$, then use L1/TV/range/optional physics losses.
Sampling	Start from $y_T \sim \mathcal{N}(0, I)$ and run DDIM/DDPM reverse steps with the same pic and x_matrix conditions. Final output is clamped to `[B,1,256,256]`.

Configuration note	Meaning
This PDF main architecture	Uses `dataset_a_256_base48.yaml`. This is the recommended 256 first-version model in the code.
Debug config difference	`dataset_a_256_debug.yaml` uses base=32, channel_mult=[1,2,4], num_res_blocks=1, cond_dim=192. Its path is the same but shorter: 256 -> 128 -> 64 bottleneck instead of 256 -> 128 -> 64 -> 32.
Base64 config difference	`dataset_a_256_base64.yaml` keeps the same topology as base48 but increases base channels to 64. Channel sequence becomes 64,128,256,256.